# Aman Pratap Singh

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## Education

**University of Maryland** College Park, MD
Master of Science in Applied Machine Learning August 2024 – May 2026

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision.

**National Institute of Technology** Hamirpur, India
Integrated M.Tech & B.Tech, Computer Science and Engineering August 2017 – May 2022

## Experience

**University of Maryland** College Park, MD
Teaching & Research Assistant | *Supervisor: Prof. Faisal Quader* August 2025 – May 2026

- Developed an **LLM/RAG-assisted anomaly detection pipeline for temporal cybersecurity streams**, combining embeddings, retrieval, dimensionality reduction, and clustering; evaluated precision & recall@5 on 5k+ events, improved suspicious-pattern triage by 13%, and mentored 30+ graduate students.

**DeepEmergence** Bengaluru, Karnataka
ML Engineer January 2023 – July 2024

- Fine-tuned **LLaMA-2 7B** on 60K financial Q&A pairs using QLoRA and DeepSpeed; improved Recall@3 from 43% to 59% on a 500-query held-out set and deployed as a FastAPI microservice on a closed-access wealth management platform.
- Architected a **RAG pipeline** grounding LLM responses in 1,200+ financial PDFs using Hugging Face embeddings, FAISS IVF-PQ, and LangChain; reduced unsupported responses by 41% (GPT-4-as-judge, 300 sampled outputs vs. non-RAG baseline) with satisfaction scores up 23% in A/B testing.
- Integrated **Azure Document Intelligence** for OCR and structured extraction from financial filings; improved extraction accuracy by 33% and cut downstream query latency by 19%.
- Built a portfolio analytics dashboard backed by a **SARIMAX-XGBoost** ensemble for 7-day return forecasting; achieved 6.2% MAPE on a 30-day rolling out-of-sample test across 50 equities, outperforming a moving-average baseline by 19%.
- Applied PCA and DBSCAN to flag irregular trading patterns; reduced false-positive rebalance signals by 17% vs. a rule-based threshold baseline, validated on 3 months of backtested data.

**Pristyn Care** Gurugram, Haryana
Data Scientist June 2022 – December 2022

- Built an **NLP chatbot** (Dialogflow) integrated with appointment and patient data systems and deployed across **WhatsApp Business API** and **Google My Business** channels; automated responses to routine query categories, reducing manual support load by 50%.
- Designed custom **ResNet** and **RNN** inspired models for patient image triage and no-show forecasting, redesigning model blocks, layer depth, and sequence flow with **87%** accuracy on 4-class severity grading; reducing scheduling backlog by **22%** across **15+** specialties.

## Projects and Research

**TerpAgent: Agentic Campus Assistant for UMD** April 2026 – Present

- Built a **FastAPI** + Pydantic backend on **Claude Sonnet 4.5** with 35 tool-use schemas across 53 endpoints for 11 UMD services; engineered an 8-round agentic loop with conversation memory and parallel tool dispatch to resolve multi-step requests in one turn.
- Designed /match endpoints returning score, why, and blockers sourced from data to prevent LLM rationale hallucination; deployed on EC2 with **Docker Compose**, regex-based heuristic fallback for 100% uptime, validated by a 13-case pytest suite and mocked-SDK agent test.

**MedVision Agent: Multimodal Diagnostic Assistant** July 2025 – Present

- Built a multimodal diagnostic assistant using **GPT-4o** and LangGraph to analyze chest X-rays with clinical notes; achieved 72% agreement with radiologist labels on a 5-class task across 500 NIH dataset samples, grounded via RAG over 100K PubMed abstracts.
- Implemented a tool-use pipeline for literature search, risk scoring, and differential diagnosis; deployed a Streamlit demo on 200+ test cases with median 3-second latency on a single GPU.

**Master Thesis – Speech Emotion Recognition** | *Advisor: Dr. Naveen Chauhan* June 2020 – June 2022

- Built a multi-speaker emotion recognition pipeline with pyannote-based diarization; extracted MFCC, Chroma, and Spectral Contrast features with augmentation for robustness on real-world dialogue audio.
- Trained **CNN-BiLSTM** models with attention on IEMOCAP and CREMA-D; achieved 83% accuracy across 6 emotion classes under 5-fold CV, outperforming SVM and Random Forest baselines by 10–12%.

## Technical Skills

**Languages:** Python, Java, SQL, C++
**AI & Generative AI:** LLMs, Hugging Face, LangChain, LangGraph, OpenAI API, RAG Pipelines, LoRA/QLoRA, FAISS (IVF-PQ)
**Machine Learning:** Scikit-learn, PyTorch, TensorFlow, Feature Engineering, Model Evaluation
**Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis
**MLOps & Cloud:** Docker, AWS (EC2, S3), FastAPI, Streamlit, MLflow, Weights & Biases